Homework 5 - 09/22 Content

MATH 1271, Garrett Credi

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Problems listed as AC are found in the Active Calculus textbook at the end of the section listed.

Problems listed as APEX are found in the APEX Calculus Textbook again at the end of the listed section.

Problems listed as PAULS-PRACT are found in Paul's Online Math Notes - Review Problems while problems listed as PAULS-ASSIGN are found in Paul's Online Math Notes - Assignment Problems and are indexed via [Topic]-[Section]-[Problem Number]

1 Type 1 - Algebraically Computing the Derivative

Choose 4 of the following problems to solve.

1. AC 1.3.3
2. AC 1.3.8
3. AC 1.3.9
4. APEX 2.1.7-14 (each of these counts as one problem)
5. PAULS-ASSIGN Derivatives.The Definition of the Derivative.1-24 (each of these counts as one problem)

2 Type 2 - Graphically Interpreting the Derivative

Choose 4 of the following problems to solve.

1. AC 1.3.2
2. AC 1.3.4
3. AC 1.3.5
4. APEX 2.1.29-32 (each of these counts as one problem)
5. APEX 2.1.33-34 (each of these counts as one problem)

3 Type 3 - Sketching Graphs

Choose 2 of the following problems to solve.

1. PAULS-ASSIGN.Derivatives.Interpretations of the Derivative.4-6 (each of these counts as one problem)
2. PAULS-ASSIGN.Derivatives.Interpretations of the Derivative.7-9 (each of these counts as one problem)

4 Type 4 - Algebraically Interpreting the Derivative

Choose 3 of the following problems to solve.

1. PAULS-ASSIGN.Derivatives.Interpretations of the Derivative.10,12,13,14,16,17,18,21 (each of these counts as one problem)

5 Type 5 - Tangent Lines

Choose 2 of the following problems to solve.

1. AC 1.3.6
2. AC 1.3.7
3. APEX 2.1.15-22 (each of these counts as one problem, ignore the bit about the "normal line")
4. PAULS-ASSIGN.Derivatives.Interpretations of the Derivative.11,15,19 (each of these counts as one problem)

6 Challenge Problem

From our discussion on the squeeze theorem we know that, for $f(x) = \sin(x)$:

$$f'(0) = \lim_{x \to 0} \frac{\sin(x)}{x} = 1 \tag{1}$$

because, when $x \approx 0$ we had that $\cos(x) \leq \dfrac{\sin(x)}{x} \leq 1$. Since this relies of $x \approx 0$, a similar strategy of using that inequality and the squeeze theorem will *not* let us compute $f'(a)$ for $a \neq 0$.

However, you *do* have the tools necessary to compute $f'(a)$ for *every* $a \neq 0$ - this is your goal for this challenge problem: to compute the derivative of $\sin(x)$ at every $x = a \neq 0$.

If you get stuck, I have placed increasingly revealing hints on the subsequent pages. I highly recommend avoiding the hints as much as possible, since I believe this is one of the most attainable results for you all to feel the feeling of "discovery."

Hint 1: You might find the

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \tag{2}$$

form of the derivative more useful here.

Hint 2: $\sin(\alpha + \beta) = \sin(\alpha)\cos(\beta) + \cos(\alpha)\sin(\beta)$.

Hint 3: Your computation of limits on worksheet 4 might be useful here.

Hint 4:

$$\frac{\cos(h)-1}{h} = -\frac{1-\cos(h)}{h^2} \cdot h \tag{3}$$